Important: Text generated by AI and may contain incorrect information, always check the content.
To analyze the main characteristics of the dataset, we'll examine its structure, data types, summary statistics, and distribution of variables.

```python
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Examine the structure of the DataFrame
display('DataFrame Shape:')
display(df.shape)

# Examine data types of the columns
display('Data Types:')
display(df.dtypes)

# Summary statistics for numerical columns
display('Summary statistics for numerical columns:')
display(df.select_dtypes(include=np.number).describe())

# Summary statistics for categorical columns
display('Summary statistics for categorical columns:')
display(df.select_dtypes(include='object').describe())

# Check the distribution of the target variable 'LUNG_CANCER'
display('Distribution of LUNG_CANCER:')
display(df['LUNG_CANCER'].value_counts())

# Check the distribution of 'GENDER'
display('Distribution of GENDER:')
display(df['GENDER'].value_counts())

# Plot histograms for numerical columns
display('Histograms for numerical columns:')
plt.figure(figsize=(12,8))
sns.histplot(df['AGE'], kde=True)
plt.title('Histogram of AGE')
plt.show()

# Check correlation between numerical variables
display('Correlation between numerical variables:')
corr_matrix = df.select_dtypes(include=np.number).corr()
plt.figure(figsize=(12,10))
sns.heatmap(corr_matrix, annot=False, cmap='coolwarm', square=True)
plt.title('Correlation Matrix')
plt.show()
```

```
'DataFrame Shape:'
(3000, 16)
'Data Types:'
```

	0
GENDER	object
AGE	int64
SMOKING	int64
YELLOW_FINGERS	int64

	0
ANXIETY	int64
PEER_PRESSURE	int64
CHRONIC_DISEASE	int64
FATIGUE	int64
ALLERGY	int64
WHEEZING	int64
ALCOHOL_CONSUMING	int64
COUGHING	int64
SHORTNESS_OF_BREATH	int64
SWALLOWING_DIFFICULTY	int64
CHEST_PAIN	int64
LUNG_CANCER	object

dtype: object

'Summary statistics for numerical columns:'

	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFF
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.0
mean	55.169000	1.491000	1.514000	1.494000	1.499000	1.509667	1.489667	1.506667	1.497333	1.491333	1.510667	1.488000	1.4
std	14.723746	0.500002	0.499887	0.500047	0.500082	0.499990	0.499977	0.500039	0.500076	0.500008	0.499970	0.499939	0.4
min	30.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
25%	42.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
50%	55.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	1.0
75%	68.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0
max	80.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0

'Summary statistics for categorical columns:'

	GENDER	LUNG_CANCER
count	3000	3000
unique	2	2
top	M	YES
freq	1514	1518

'Distribution of LUNG_CANCER:'

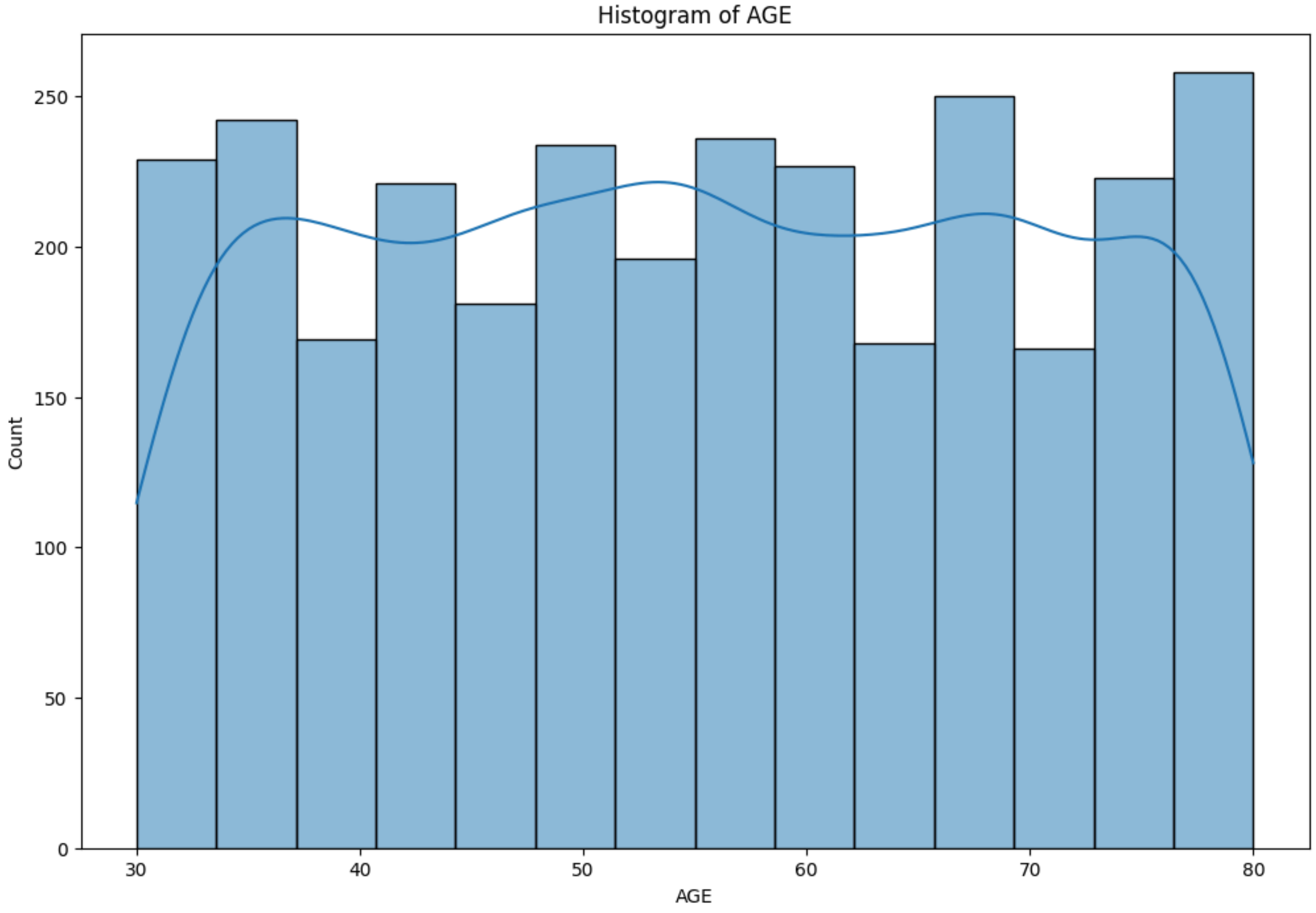
	LUNG_CANCER
YES	1518

	LUNG_CANCER
NO	1482

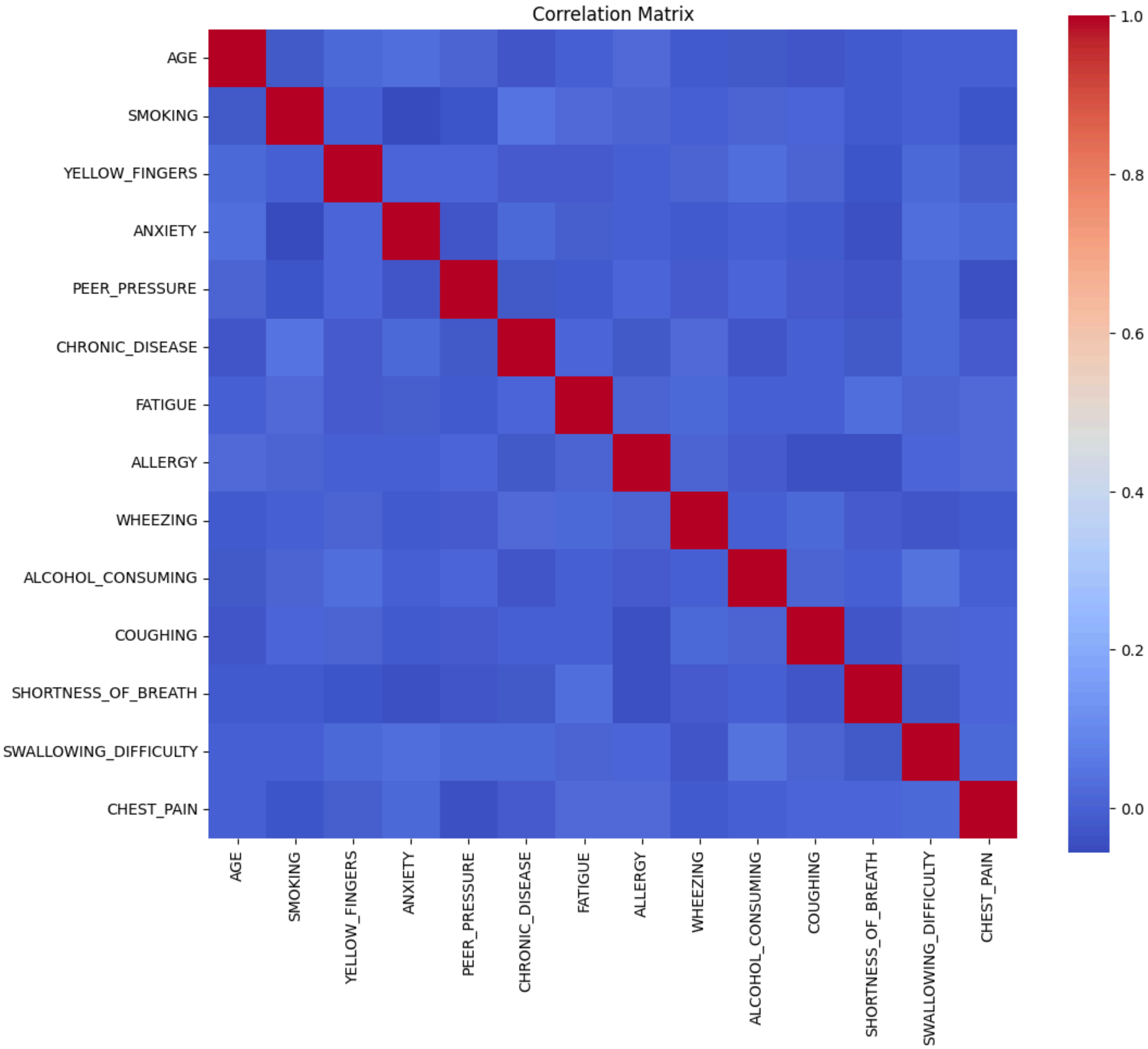
dtype: int64
'Distribution of GENDER:'

	GENDER
M	1514
F	1486

dtype: int64
'Histograms for numerical columns:'



'Correlation between numerical variables:'




```
The analysis is focused on understanding the characteristics of the dataframe.
```